

Hugo C. Sámano-Sánchez

Curriculum Vitae

✉ samano@intl.zju.edu.cn

person.zju.edu.cn/samano

[hugocarlos.github.io](https://github.com/hugocarlos)

Google Scholar: [c5EwcVAAAAAJ](https://scholar.google.com/citations?user=c5EwcVAAAAAJ)

ORCID: [0000-0003-4744-4787](https://orcid.org/0000-0003-4744-4787)

Summary

My research integrates evolutionary biology, structural bioinformatics, and AI-driven approaches to decode protein-protein interactions and mechanisms of pathogenesis. Trained in computational biology and infection biology, my current areas of interest combine the development of algorithms for infection diagnosis, understanding the evolutionary constraints and protein folding principles that shape signaling complexes, and predicting novel mechanisms of pathogenesis. In my recent independent research group at the Zhejiang University-University of Edinburgh Institute, I lead projects that bridge fundamental evolutionary questions with translational applications in disease. I am eager to collaborate with groups at the intersection of cell signaling, infection, and big data to advance both basic research and clinical outcomes.

Appointments

- 09/25 - current **ZJE Programme Co-ordination Group member**, Zhejiang University-University of Edinburgh Institute, Haining, China
- 10/24 - current **Assistant Professor, tenure track**, Zhejiang University-University of Edinburgh Institute, Haining, China
- 05/22 - current **PhD Supervisor**, *College of Medicine and Veterinary Medicine*, University of Edinburgh, Scotland
- 09/21 - current **Honorary Lecturer**, *College of Medicine and Veterinary Medicine*, University of Edinburgh, Scotland

Education

- 10/15 - 12/19 **Ph.D. in Molecular Biology**, European Molecular Biology Laboratory / University of Heidelberg, Heidelberg, Germany
Magna cum laude.
- 09/13 - 02/15 **Joint Master Programme in Infectious Diseases, Vaccinology and Drug Discovery**, University of Basel, Switzerland and the National University of Singapore, Singapore
- 08/07 - 06/11 **B.Sc. in Genomics Sciences**, National Autonomous University of Mexico, Cuernavaca, Mexico
With honours.

Professional Experience

- 10/24 - current **Assistant Professor**, Zhejiang University-University of Edinburgh Institute, Haining, China
- 05/20 - 09/24 **Lecturer**, *Biomedical Informatics bachelor's programme*, Zhejiang University-University of Edinburgh Institute, Haining, China
- 10/15 - 01/20 **PhD Student / Bridging Postdoctoral Fellow**, European Molecular Biology Laboratory, Heidelberg, Germany
- Advisor: Dr. Toby J. Gibson.
 - Eukaryotic Short Linear Motifs, bacterial mimics, *Plasmodium*-Host protein interactions.
- 01/14 - 02/15 **Master thesis**, Vivax Malaria Lab, National University of Singapore, Singapore
- Advisors: Dr. Bruce M. Russell and Prof. Dr. Markus R. Wenk.
 - Lipidomics, software development, protein modeling.

- 10/12 - 07/13 **Research Assistant**, National Institute of Public Health, Cuernavaca, Mexico
- Advisor: Dr. Jesus Martinez-Barnetche.
 - Lymphocyte antigen receptor analysis.
- 01/12 - 09/12 **Bioinformatics Consultant**, Winter Genomics (<http://www.wintergenomics.com>), Cuernavaca, Mexico

Scientific Publications

* Contributed equally.

- Sanogo, B., Echoru, I., Jones, C., Butala, C., Lin, D., Maiga, H., **Sámano-Sánchez, H.**, Sacko, M., Kassegne, K., Welburn, S.C. A reservoir at the gates: a scoping review of nonhuman mammalian hosts for human schistosomiasis in western Africa and the critical challenge for elimination. *Infectious Diseases of Poverty*, Accepted.
- Giménez, A., Del Giudice, M., López, P., Guaimas, F., **Sámano-Sánchez, H.**, Gibson, T., Chemes, L., Arregui, C., Ugalde, J., Czibener, C. Brucella NpeA is a secreted Type IV effector containing an N-WASP binding short linear motif that promotes niche formation. *mBio*, June, 2024. doi: 10.1128/mbio.00726-24
- **Sámano-Sánchez, H.**, Gibson, T.J., Chemes, L.B. Using Linear Motif Database Resources to Identify SH2 Domain Binders. *Methods Mol Biol*, 2023. doi: 10.1007/978-1-0716-3393-9_9
- Kumar, K., Michael, S., Alvarado-Valverde, J., Meszaros, B., **Sámano-Sánchez, H.**, Zeke, A., Dobson, L., Lazar, T., Ord, M., Nagpal, A., Farahi, N., Kaser, M., Kraleti, R., Davey, N.E., Pancsa, R., Chemes, L.B., Gibson, T.J. The eukaryotic linear motif resource: 2022 release. *Nucleic Acids Research*, 2022. doi: 10.1093/nar/gkab975
- Mezsaros, B., **Sámano-Sánchez, H.**, Alvarado-Valverde, J., Calyseva, J., Martinez-Perez, E., Alves, R., Shields, D.C., Kumar, M., Rippmann, F., Chemes, L.B., Gibson, T.J. Short linear motif candidates in the cell entry system used by SARS-CoV-2 and their potential therapeutic implications. *Science Signaling*, 2021. doi: 10.1126/scisignal.abd0334
- **Sámano-Sánchez, H.**, Gibson, T.J. Mimicry of Short Linear Motifs by Bacterial Pathogens: A Drugging Opportunity. *Trends in Biochemical Sciences*, 2020. doi: 10.1016/j.tibs.2020.03.003
- Kumar, M., Gouw, M., **Sámano-Sánchez, H.**, Pancsa, R., Glavina, J., Diakogianni, A., Alvarado-Valverde, J., Bukirova, D., Calyseva, J., Palopoli, N., Davey, N.E., Chemes, L.B., Gibson, T.J. ELM - the eukaryotic linear motif resource in 2020. *Nucleic Acids Research*, 2019. doi: 10.1093/nar/gkz1030
- Sampietro, D., **Sámano-Sánchez, H.**, Davey, N.E., Sharan, M., Meszaros, B., Gibson, T.J., Kumar, M. Conserved SQ and QS motifs in bacterial effectors suggest pathogen interplay with the ATM kinase family during infection. *bioRxiv*, 2018. doi: 10.1101/364117
- Gouw, M., Michael, S., **Sámano-Sánchez, H.**, Kumar, M., Zeke, A., Lang, B., Bely B., Chemes, L.B., Davey, N.E., Deng, Z., Diella, F., Gurth, C.-M., Huber, A.-K., Kleinsorg, S., Schlegel, L.S., Palopoli, N., Roey, K.V., Altenberg, B., Remenyi, A., Dinkel, H., Gibson, J.T. The eukaryotic linear motif resource - 2018 update. *Nucleic Acids Research*. doi: 10.1093/nar/gkx1077
- Kosaisavee, V., Suwanarusk, R., Chua, A.C.Y., Kyle, D.E., Malleret, B., Zhang, R., Imwong, M., Imerbsin, R., Ubalee, R., **Sámano-Sánchez, H.**, Yeung, B.K.S., Ong, J.J.Y., Lombardini, E., Nosten, F., Tan, K.S.W., Bifani, P., Snounou, G., Rénia, L., Russell, B. Strict tropism for CD71+/CD234+ human reticulocytes limits the zoonotic potential of *Plasmodium cynomolgi*. *Blood*, 2017. doi: 10.1182/blood-2017-02-764787
- Gouw, M., **Sámano-Sánchez, H.**, Roey, K.V., Diella, F., Gibson, T.J., Dinkel, H. Exploring Short Linear Motifs Using the ELM Database and Tools. *Current Protocols in Bioinformatics*, 2017. doi: 10.1002/cpbi.26
- Godoy-Lozano, E.E., Tellez-Sosa, J., Sanchez-Gonzalez, G., **Sámano-Sánchez, H.**, Aguilar-Salgado, A., Salinas-Rodriguez, A., Cortina-Ceballos, B., Vivanco-Cid, H., Hernandez-Flores, K., Pfaff, J.M., Kahle, K.M., Doranz, B.J., Gomez-Barreto, R.E., Valdovinos-Torres, H., Lopez-Martinez, I., Rodriguez, M.H., Martínez-Barnetche, J. Lower IgG somatic hypermutation rates during acute dengue virus infection is compatible with a germinal center-independent B cell response. *Genome Medicine*, 2016. doi: 10.1186/s13073-016-0276-1
- Cortina-Ceballos, B., Godoy-Lozano, E.E., Tellez-Sosa, M., Ovilla-Munoz, M., **Sámano-Sánchez, H.**, Aguilar-Salgado, A., Gomez-Barreto, R.E., Valdovinos-Torres, H., Lopez-Martinez, I., Aparicio-Antonio, R., Rodriguez, M.H., Martínez-Barnetche, J. Longitudinal analysis of the peripheral B cell repertoire reveals unique effects of immunization with a new Influenza virus strain. *Genome Medicine*, 2015. doi: 10.1186/s13073-015-0239-y
- Cortina-Ceballos, B.*, Godoy-Lozano, E.E.*, **Sámano-Sánchez, H.***, Aguilar-Salgado, A., Velasco-Herrera, M.D.C., Vargas-Chavez, C., Velazquez-Ramirez, D., Romero, G., Moreno, J., Tellez-Sosa, J., Martínez-Barnetche, J. Reconstructing and mining the B cell repertoire with ImmuneDiversity. *mAbs*, 2015. doi: 10.1080/19420862.2015.1026502
- Rosenblueth, M., Sayavedra, L., **Sámano-Sánchez, H.**, Roth, A., Martínez-Romero, E. Evolutionary Relationships of Flavobacteria and Enterobacterial Endosymbionts with their Scale Insect Hosts (Hemiptera: Coccoidea). *Journal of Evolutionary Biology*, 2011. doi: 10.1111/j.1420-9101.2012.02611.x

Awards and Scholarships

- Zhejiang University Teaching Achievement Award (2025).
- Second prize at the Teaching Competition for young faculty of Zhejiang University-University of Edinburgh Institute (2022).

Talks

Invited talks

- Talk for the BioLinkX seminar series, Zhejiang University Student Bioinformatics Association, Hangzhou, China, 2023.
- Talk to the Parasitology Unit of the Center for Infectious Diseases, Heidelberg University Hospital, Heidelberg, Germany, 2019.
- Institutional seminar at the Biotechnology Research Institute (IIB), National University of San Martin (UNSAM), Buenos Aires, Argentina, 2017.

Selected talks

- Cold Spring Harbor Asia Conference: Bacterial Infection and Host Defense, Suzhou, China, 2019.
- VIII Argentinian Bioinformatics and Computational Biology Congress, Posadas, Argentina, 2017.
- 5th International Conference of Research on *Plasmodium vivax* Malaria, Bali, Indonesia, 2015.

Teaching

Mentor

- PhD supervisor (three students), dual degree ZJU-UoE Integrative Biomedical Sciences programme, currently.
- Master thesis supervisor (one student), University of Edinburgh Biomedical Sciences programme, 2023.

Course Organizer

- Biomedical Informatics 3 (BMI3), Biomedical Informatics Programme, Zhejiang University-University of Edinburgh Institute, China, 2023 to 2025. This is a core course on algorithm design.
- Integrative Biomedical Sciences 3 (IBMS3), Biomedical Informatics and Biomedical Sciences Programmes, Zhejiang University-University of Edinburgh Institute, China, 2022. This is a core course on career development and experimental design.
- Bioinformatics Tools for Protein Structure, Disorder and Interaction Analysis, National University of San Martin, Buenos Aires, Argentina, 2017.

Lecturer

- Linear Algebra (AM1), Introduction to Biomedical Informatics 1 (IBI1), Genomics and Proteomics 2 (GP2), Infection 3 (IN3), Computational Biology and Systems Biology 3 (CBSB3) and Computational Modeling and Machine Learning (CMML3) courses of the Biomedical Informatics and Biomedical Sciences Programmes, Zhejiang University-University of Edinburgh Institute, China, 2020-2025.
- Biomedical Disorders course of the Integrative Biomedical Sciences PhD programme, Zhejiang University-University of Edinburgh Institute, China, 2021-2025.
- Biomedical Frontiers course of the Artificial Intelligence and Digital Health MSc. programme, Zhejiang University, China, 2023.
- Basic Discrete Mathematics. Zhejiang University-University of Illinois Urbana-Champaign (ZJU-UIUC) Institute, China, 2024.

Skills and Interests

Languages Spanish (native), English (advanced), French (basic), Chinese (basic)

Research interests Host-Pathogen interactions. Evolution. Protein Modeling. Algorithm development.